

Weekly Research Summary — May 29, 2026

Adversarial Fairness Attacks on DRO-FAIR

Both tasks assigned by Madam are COMPLETE

Task 1: Tabular Data (270 experiments)

Dataset	Attack	α	DRO Wins By	p-value
Credit	IF	0.2	64.5%	0.031*
Credit	IF	0.3	97.5%	0.031*
LSAC	IF	0.3	96.2%	0.031*

DRO-FAIR significantly reduces DP violation under IF attacks
(Wilcoxon signed-rank, $n=5$ seeds)

Task 2: UTKFace Images (9 experiments)

α	Condition	Naive DP	DRO DP	Winner
0.0	Clean	0.036	0.027	DRO ✓
0.1	Corrupted	0.094	0.130	Naive ✓
0.2	Corrupted	0.103	0.110	Naive ✓

NEW FINDING: DRO makes fairness WORSE on image data!
Opposite of tabular results.

KEY INSIGHT: DRO-FAIR robustness is NOT universal

- Metric-dependent: DRO wins against IF attacks → loses against DP attacks
- Modality-dependent: DRO wins on tabular data → loses on image features

This is a NOVEL finding – prior work only tested random noise, not gradient-based adversarial attacks.

Next: More UTKFace seeds → CelebA/FairFace → ResNet50 → Paper draft (NeurIPS/ICLR)